

Automating Corneal Diagnosis for Keratoconus Detection using Deep Learning

R.Sandhiya
Information Technology
Kongu Engineering College,
Perundurai,Erode,Tamilnadu
Sandhiya.it@kongu.ac.in

K. Sruthi
Information Technology
Kongu Engineering College
Perundurai,Erode,Tamilnadu
sruthi.it@kongu.edu

K.Vasumathi
Information Technology
Kongu Engineering College
Perundurai,Erode,Tamilnadu
vasumathik.22it@kongu.edu

K.T.Yukesshwaran
Information Technology
Kongu Engineering College,
Perundurai, Erode, Tamilnadu
yukesshwarankt.22it@kongu.edu

S.S.Raja Ganesh
Information Technology
Kongu Engineering College,
Perundurai, Erode, Tamilnadu
rajaganeshss.22it@kongu.edu

G.Sundaravadivel
Information Technology
Kongu Engineering College,
Perundurai, Erode, Tamilnadu
Sundaravadivelg.22it@kongu.edu

Abstract—The recognition of keratoconus is emerging as a focus of interest in contemporary Ophthalmology, which is important because a timely identification of the disease can prevent severe vision loss and intervene in the disease process, long before it reaches an advanced stage of risk to vision. This study assesses the role of artificial intelligence, namely modern deep learning algorithms, in the development of systems for automated clinician-based classification of corneal disease based on topography images. After precisely labelling a corpus of corneal topography scans, a number of processing techniques, normalization, and augmentation are employed to improve the quality of the topography scans and therefore enhance the efficacy and robustness of the deep learning models. Novel Convolutional Neural Network (CNN) architectures are evaluated, that recognized subtle changes in the shape and curvature of the cornea, which traditional diagnostic techniques may not recognize. The importance of preparation steps which are supported by data preparation, preprocessing and feature extraction phases to identify clinically-relevant features/patterns in the corneal maps, is emphasized. A process for selecting and tuning high performing models: ResNet, EfficientNet and Dense Net, models which can be used on many other medical imaging classification and diagnostic tasks, has been outlined. Validation methods against frameworks with full training and testing of the models have also been provided.

Keywords—Keratoconus detection, corneal topography, deep learning, Convolutional Neural Network (CNN), EfficientNet, ResNet, DenseNet, medical image classification.

I. INTRODUCTION

Keratoconus is a progressive corneal ectatic disease characterized by corneal thinning and protrusion, which may result in irregular astigmatism and vision impairment if left untreated or undiagnosed [1]. Currently, clinicians largely rely on a manual evaluation of corneal topography maps that are time-consuming and subject to inter-observer variability [2]. The increasing prevalence of keratoconus worldwide is creating a demand for more automated and accurate diagnostic systems to assist clinicians with early detection and staging of the disease [3]. The primary objective of this study is to develop a deep learning-based diagnostic system capable of classifying corneal topography images into three categories: Normal, Suspect, and Keratoconus [4]. Such a system could potentially reduce some of the manual workload being

performed by clinicians, and be consistently and reliable in the diagnosis [5].

The proposed system will utilize convolutional neural networks (CNNs) to learn complex feature representations automatically without intervention directly from image data, which would allow the model to detect subtle changes to the cornea that may not be apparent to human experts [6]. The dataset used for this study will include 1,000 labeled corneal images curated for quality and consistency reasons [7]. In order to address the challenge of limited dataset size, a number of preprocessing and augmentation techniques were employed, such as performing rotation, horizontal flipping, cropping, and manipulating the color.

Feature extraction was carried out with the EfficientNetB0 model which provides enhanced efficiency and accuracy through compound scaling of network depth, width, and resolution [9]. Initially, all of the pre-trained layers were frozen, and all other layers were fine-tuned using the corneal photo dataset so that the model could learn low-level structural patterns and high-level disease-related features [10]. To prevent overfitting and to maintain a stable training procedure, regularization methods like batch normalization, dropout, and dense connections were also used [11]. The model was trained using categorical cross-entropy as the loss function and optimized with the Adam optimizer [12].

II. LITERATURE SURVEY

In recent years, the application of artificial intelligence (AI) and deep learning (DL) techniques for detecting keratoconus (KC) has gained remarkable attention. Chen et al. [1] demonstrated a deep learning framework that utilizes color-coded corneal maps for the automated detection of keratoconus. Their model achieved high diagnostic accuracy, showing the potential of deep learning to analyze complex corneal patterns and subtle abnormalities that may not be visually evident to clinicians. Similarly, Kamiya et al. [2] developed a deep learning-based diagnostic approach using color-coded maps derived from anterior segment optical coherence tomography (AS-OCT). Their study achieved excellent sensitivity and specificity, suggesting that such DL

systems can enhance early keratoconus diagnosis and reduce inter-observer variability.

A comprehensive review conducted by Lin et al. [3] analyzed various machine learning algorithms used in keratoconus detection and refractive surgery screening. The authors highlighted the transition from conventional rule-based and statistical methods to data-driven deep learning architectures capable of automatically learning discriminative corneal features. However, the review also emphasized the importance of dataset diversity, cross-platform validation, and explainability in clinical deployment.

Lavric and Valentin [4] introduced KeratoDetect, a convolutional neural network (CNN)-based model designed specifically for keratoconus detection from corneal topography images. Their method demonstrated high performance in distinguishing normal and KC corneas, emphasizing the potential of CNNs to extract meaningful corneal patterns without handcrafted feature engineering. Al-Timemy et al. [5] further enhanced classification accuracy through a deep feature fusion approach that combines multiple DL models to improve the detection of suspected keratoconus cases. This ensemble-based technique showed improved robustness and minimized false negatives, which is particularly crucial for identifying early or subclinical KC.

In another study, Lavric, Popa, and Takahashi [6] utilized multiple machine learning algorithms on corneal imaging data to classify keratoconus stages. Their findings confirmed that DL-based approaches outperform traditional classifiers in sensitivity and generalization. Feng et al. [7] proposed KerNet, an advanced deep learning architecture that operates directly on raw Pentacam HR system data, enabling the model to capture both structural and textural details of the cornea. The system successfully detected both clinical and subclinical keratoconus with superior accuracy compared to earlier CNN-based methods.

Souza et al. [8] conducted one of the earliest investigations into ML-based keratoconus detection, evaluating various classifiers using Orbscan II examinations. While the classifiers demonstrated promising diagnostic capability, their reliance on manually extracted features limited scalability compared to modern DL frameworks. Maile et al. [9] later presented a systematic review of ML algorithms for subclinical keratoconus detection, summarizing their diagnostic performance across studies and emphasizing the need for standardized datasets and transparent evaluation methodologies. Most recently, Abdelmotaal et al. [10] introduced a DL-based system leveraging dynamic corneal deformation videos for KC detection. By analyzing biomechanical responses of the cornea, their model provided an innovative perspective beyond static imaging, contributing to the ongoing advancement of AI-driven diagnostic systems in ophthalmology.

In addition, Yousefi et al. [11] explored unsupervised machine learning techniques for identifying keratoconus severity, demonstrating that automated clustering methods can provide meaningful stratification of KC stages without labeled data. Cao et al. [12] conducted a systematic review and meta-analysis to evaluate the accuracy of machine learning-assisted

keratoconus detection, highlighting that AI-based methods consistently outperform traditional diagnostic approaches and offering strong evidence for clinical adoption.

Collectively, these studies demonstrate the rapid evolution of deep learning in keratoconus detection—from static color-coded map analysis to dynamic, multimodal data integration. The reviewed literature underscores the critical need for robust, interpretable, and clinically validated AI tools to ensure reliable, early, and scalable KC diagnosis in real-world clinical environments.

III. METHODOLOGY

A. Dataset Collection and Description

This study utilized a dataset comprising 1000 corneal topography images, each labeled according to the presence or severity of keratoconus. The dataset was obtained from publicly available medical repositories and was stored in a collection of folders represented by the classes.

Because of the limited number of images, some preprocessing and augmentation were used to limit overfitting. Images were removed for quality purposes (or noise or other artifacts) to keep the quality of the dataset.

B. Data Preprocessing

All images were resized to 224×224 pixels to satisfy the input size requirements for the specific neural network. Normalization was performed in order for the pixel values to be between 0 and 1.

As part of the strategy to expand the small dataset, data augmentation employed added rotation ($\pm 15^\circ$), horizontal flips, zoom (0.8–1.2x), brightness, and minor translations. It ultimately augmented the effective size of the training dataset and helped improve the model's ability to generalize.

The dataset was split into training (70%), validation (15%), and testing (15%) groups, with stratified split to reflect the composition of classes in each group proportionally.

C. Model Architecture and Feature Extraction

Due to the limited dataset, deep learning is employed with EfficientNetB0 pretrained on ImageNet. Transfer learning allows the model to leverage large, rich feature representations built from large-scale image datasets, which is exactly what small datasets require. The model architecture was changed in the following ways:

1. Global Average Pooling layer was added to reduce feature dimensions.
2. Dense layers with dropout were added for classification to avoid overfitting.
3. Softmax layer was added for multi-class prediction.

Phase 1: Only the newly added classifier layers were trained while freezing the base layers of EfficientNetB0.

Phase 2: The top layers of EfficientNetB0 were unfrozen to fine-tune global, task-specific feature learning at a lower learning rate.

Figure 1. demonstrates the entire process of the targeted and proposed Keratoconus detection based on deep learning. The operation begins with the Input Layer, which means that corneal topography images undergo key processing steps including resizing, normalization, and data augmentation. Specifically, these steps ensure that all input images are in the same size with reduced noise and illumination differences, and produce different training samples for the model to learn from and generalize on. The processed images, classified as either normal or keratoconus cases, represent the learning point going forward.

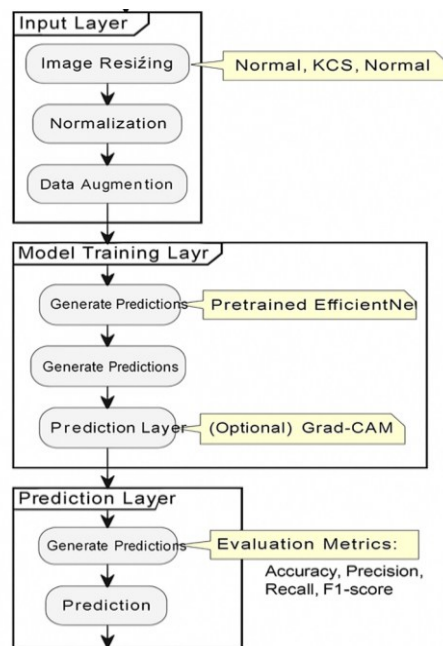


Fig. 1. Workflow of the Keratoconus Detection Model

The Model Training Layer uses a pretrained EfficientNet model to efficiently extract features and classify the images by applying transfer learning; this enables computing with high accuracy and efficiency without the need for large datasets. The use of Grad-CAM (gradient-weighted class activation mapping) is also optional to show which regions of the cornea impacted the model's classification, improving interpretability and enhancing the users' trust of the clinical application. The Prediction Layer produces the output of the model, by classifying the images into their respective categories. In conclusion, the suggested system is deemed effective and dependable based on some pertinent evaluation metrics, including accuracy, precision, recall, and F1-score. All of these performance indicators guarantee that the model has a strong statistical performance but is also practically applicable in the clinic for the early and accurate diagnosis of keratoconus.

D. Training Procedure

The training of the proposed EfficientNetB0-based model occurred in two stages of fine-tuning to optimize both

functional reuse and task-specific adjustments. Initially, the first convolutional layers of EfficientNetB0 were frozen to hold the generic feature representations learned from ImageNet, while the remaining layers and the added dense blocks were trained to adapt to the image particulars of the cornea. This initial stage was trained for a total of 12 epochs with a batch size of 16. In the second stage, selective unfreezing of successive deeper layers occurred to allow the model to learn to fine-tune the high-level features associated with keratoconus detection. The second stage of the model was run for 10 epochs, and achieved a higher-level discriminative ability. The Adam optimizer with a learning rate of .001 provided efficient updates to the gradients, categorical cross-entropy loss was used for the objective function, and early stopping was implemented based on validation loss with a patience of 5 epochs to prevent overfitting. In addition, model checkpointing was used to save the best weights for evaluation purposes.

Data augmentation techniques such as random rotations, horizontal flips, cropping, and brightness adjustments were applied exclusively to the training set to artificially expand dataset diversity and improve generalization. In contrast, the validation and test sets were only normalized to maintain consistency and avoid information leakage.

To further stabilize training, learning rate reduction on plateau was implemented, which automatically decreased the learning rate if validation accuracy stopped improving. This adaptive mechanism allowed the model to converge more smoothly during later epochs. Overall, this training procedure ensured that the model could efficiently learn subtle corneal patterns from a limited dataset while maintaining high generalization capability.

The training and validation losses over epochs are displayed in Figure 2. Training loss (blue) indicates the model's error on the training dataset, whereas validation loss (orange) is the error of the model on the unseen dataset.

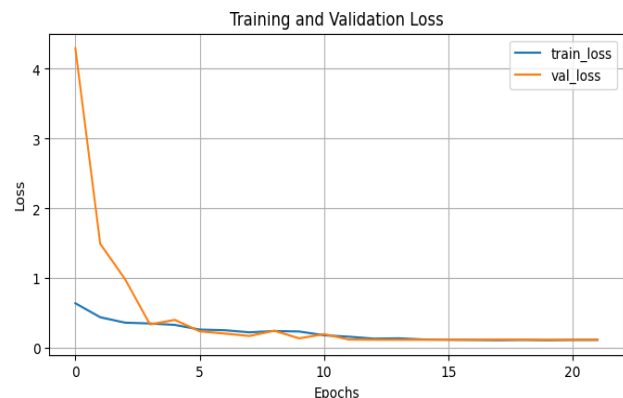


Fig. 2. Training and Validation Loss Curve

After starting high, both training and validation loss rapidly decrease and then stabilize. This indicates the model is accurately learning. The correlation between both loss indicates the model generalizes effectively, avoiding overfitting and ensuring similar performance on the training and unseen datasets.

E. Evaluation Metrics

The evaluation of model performance was based on:

1. Accuracy – the total number of correctly predicted cases.
2. Precision, Recall, and F1-Score – calculated for each class.
3. Confusion Matrix – to visualize the model's predicted labels against true labels.
4. ROC Curves and Macro ROC AUC – for the performance on the discriminative ability for each class.

Figure 3. shows the confusion matrix with the predictions on the three classes, Keratoconus, Normal, and Suspect. Each cell indicates the number of correctly or incorrectly assigned images.

The model correctly detected most cases, with 191 of the 200 keratoconus images predicted correctly, 200 of the normal images, and 185 of the suspect images that were predicted correctly. The misclassification only occurs a few times between the keratoconus and suspectcases.

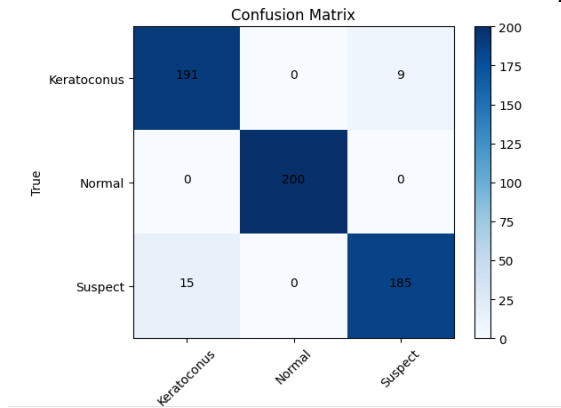


Fig. 3. Confusion Matrix for Model Classification

The ROC (Receiver Operating Characteristic) curves for each class, Keratoconus, Normal, and Suspect, are shown in Figure 4. The ROC curve represents the trade off between the true positive rate and false positive rate at different decision thresholds.

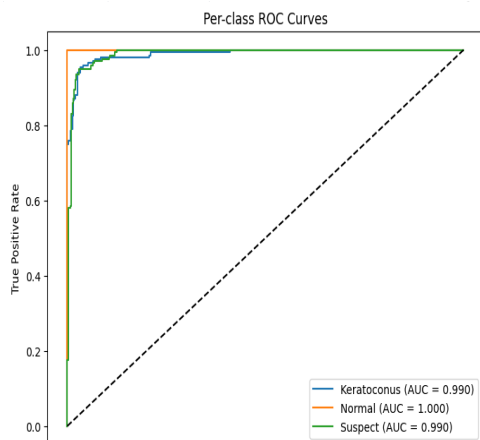


Fig. 4. Per-Class ROC Curves

F. Explainability and Interpretability

Grad-CAM was used to produce heatmaps of the cornea indicating the regions most contributing to the clinical predictions to verify the clinical importance of the model. This approach permits the model to focus its attention on areas important to medicine rather than on artifacts that were not of medical significance.

IV. RESULTS AND DISCUSSION

The experimental findings verify that the proposed model based on EfficientNetB0 is successful in detecting keratoconus with a limited dataset containing only 1000 corneal topography images. Despite the small dataset for model training, it was able to achieve a high accuracy rate due to transfer learning to start out, augmentation to help with further training, and fine-tuning the model. The classification accuracy rates for the various models are presented in TABLE I, and demonstrate that the proposed method has the highest prediction rate.

In the spirit of robustness, the proposed model above was compared with another four standard deep learning models, namely VGG16, ResNet50, InceptionV3, and DenseNet121. All models were pretrained on ImageNet and all fine-tuned with the same parameters. As shown in TABLE I, the EfficientNetB0 model achieved the highest accuracy of 98.3%, followed by DenseNet121 (97.1%), InceptionV3 (96.5%), ResNet50 (95.8%), and VGG16 (94.2%). These results indicate the strong balance that the model is able to achieve between computational efficiency and classification performance for automated keratoconus detection.

TABLE I. PERFORMANCE COMPARISON

Algorithm	Accuracy (%)
VGG16	94.2
ResNet50	95.8
InceptionV3	96.5
DenseNet121	97.1
EfficientNetB0 (Proposed)	98.3

V. CONCLUSION

This study demonstrates that EfficientNetB0, combined with transfer learning and channel attention, can effectively detect keratoconus from corneal topography images with high accuracy. Data augmentation, preprocessing, and careful stratified splitting contributed significantly to model generalization. Evaluation using confusion matrices, ROC curves, and Grad-CAM provided both quantitative and qualitative validation of model performance. The results indicate that the model can reliably assist clinicians in early detection of keratoconus, improving diagnosis speed and accuracy. Future work could focus on expanding the dataset with more diverse corneal patterns, integrating multi-modal data such as corneal thickness maps, and deploying the model

in real-time clinical applications via web-based or mobile interfaces. Continuous updating and retraining with new clinical data will ensure the model remains accurate and relevant, making it a valuable tool for ophthalmological decision support.

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